Class 5: Integrated Sensing and Communication

Waveforms, Resource Allocation, Attacks, Prototypes, Standards

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CIS 4930/5930 Future Edge Networks
xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks

Two machines, one sky

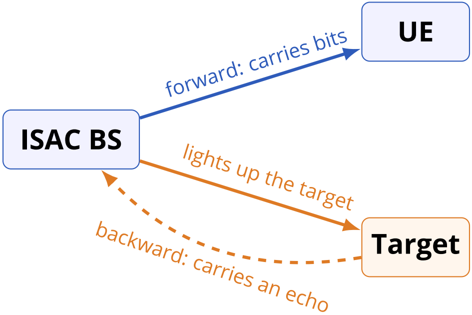
Frankfurt airport surveillance radar

Ericsson active 5G antennas

Q: One sends a wave to be understood. One sends a wave to come back. What do they share?

A: Spectrum, power, antennas, and silicon — all of it scarce. Building two systems that never talk to each other pays for the same hardware twice.

One waveform, two jobs



The same transmission is information on the way out and measurement on the way back.

ISAC does not add a radar to the network.
It reads the network's own signal as radar.

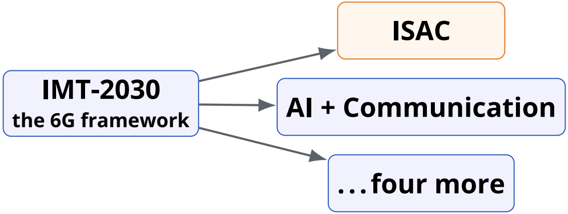
An echo is data nobody uploaded


Waymo Jaguar I-Pace roof sensor package

- **Vehicles:** see what the camera cannot.
- **XR:** track a hand, no wearable.
- **IoT:** count people, no camera.
- **Space:** range the satellite you talk to.
- **Safety:** a drone, a fall, a flood.
- **Farming, weather, health:** same echo.

Every one of these is the network noticing something on its own.

XR = Extended Reality   IoT = Internet of Things



Q: Why does a standards body have to say this out loud?

A: Because a usage scenario is what gets spectrum, performance requirements, and test procedures written for it. Until ISAC was named, no radio was obliged to be measurable.

ITU-R Recommendation M.2160, IMT-2030 framework (2023)

Two boxes became one box

Separate systems

Comms — own band, own antenna

Radar — own band, own antenna

DFRC

One transmit chain — bits out, echo in

What it buys

One spectrum licence, one amplifier, one aperture, one power budget.

DFRC = Dual-Function Radar Communication

What it costs

A single waveform now has to satisfy two specifications at once.

DFRC is one box. ISAC is the question.

ISAC

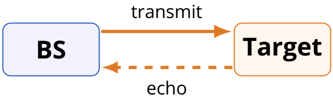
DFRC
one shared box

separate devices,
separate sites,
one sensing picture

DFRC shares waveforms, antennas, and processing chains inside one radio.

ISAC also lets one node transmit and a different node listen.

Every DFRC system is ISAC.
Not every ISAC system fits in one box.



- **Delay** $\rightarrow$ range, $R = c\tau/2$
- **Doppler** $\rightarrow$ closing speed
- **Angle** $\rightarrow$ direction

Q: Which knob buys finer range, and which buys finer speed?

A: Bandwidth buys range resolution — a wider signal has a sharper peak in time. **Observation time** buys velocity resolution — you must watch long enough for the phase to drift.

The four hard parts, named up front

Techniques

- Waveform design
- MIMO, beamforming
- Shared RF hardware
- Coding and allocation

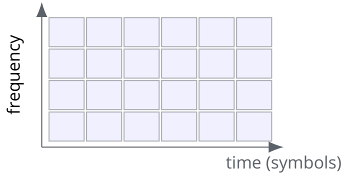
Challenges

- Tradeoff: unavoidable
- Interference: mutual
- Complexity: non-convex
- Regulation: who owns it

MIMO = Multiple Input Multiple Output RF = Radio Frequency

Three of the four are engineering.
The fourth is a committee.

OFDM already carries a radar inside it

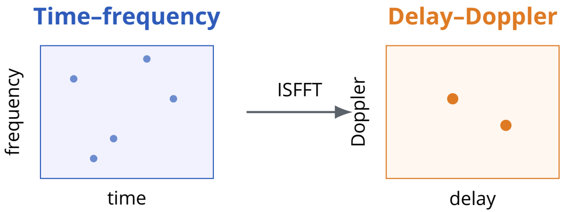


- Mature: Wi-Fi, 4G, 5G, 6G.
- The grid is already knobs.
- A cell serves bits or echoes.
- No new modulator needed.

Q: If OFDM is good enough, why design new waveforms?

A: Because OFDM is described in time and frequency, and a moving target is naturally described in delay and Doppler. Fast targets smear an OFDM grid; they sit still in a delay–Doppler one.

OFDM = Orthogonal Frequency Division Multiplexing



A target that smears across time and frequency is a point in delay and Doppler.

ISFFT = Inverse Symplectic Finite Fourier Transform

Three waveforms, three bets

	Symbols live in	Strength	Weakness
OFDM	Time–frequency	Deployed everywhere; subcarriers double as radar bins	Doppler breaks subcarrier orthogonality
OTFS	Delay–Doppler, mapped up by ISFFT	Robust to high mobility; simple to bolt onto OFDM	Leakage between bins; high PAPR
ODDM	Delay–Doppler, natively	Truly orthogonal pulses; sparse, clean echoes	Early research; hardware immature

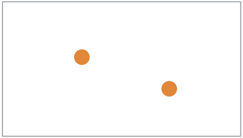
OTFS = Orthogonal Time Frequency Space   ODDM = Orthogonal Delay-Doppler Division Multiplexing
PAPR = Peak-to-Average Power Ratio, which decides how hard the amplifier works

OFDM is what ships.
ODDM is what the sensing side wants.

Why orthogonality is worth the trouble

Leaky basis (OTFS)

Orthogonal basis (ODDM)



Q: Two blurred dots or two sharp dots — why does the estimator care?

A: Leakage is energy in the wrong bin. It raises the sidelobes, hides a weak target next to a strong one, and shows up as interference between users.

They want opposite things (1/2)

	Sensing wants	Communication wants
Spectrum	Wide bandwidth, for resolution	Efficient bandwidth, for rate
Signal	Deterministic waveforms	Random, information-bearing symbols
Power	High transmit power on the target	Power efficiency per bit
Time	Continuous illumination	Low-latency bursts
Space	Wide or scanning beams	Narrow beams, one per user

Randomness carries information and ruins an autocorrelation.
That single sentence is the whole conflict.

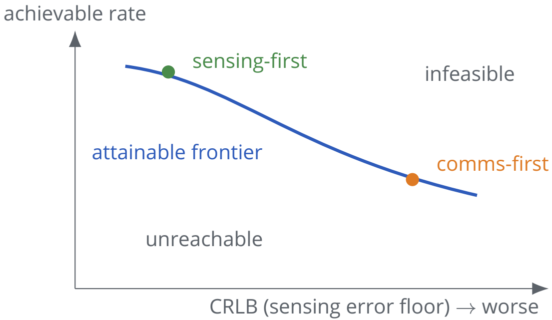
They want opposite things (2/2)

	Sensing wants	Communication wants
Interference	A clean echo, no clutter	A clean constellation, no radar leakage
Domain	Delay–Doppler resolution	Time–frequency multiplexing
Hardware	Precise timing, high-power amplifiers	Spectral efficiency, many users
Latency	Low-latency tracking feedback	Tolerates delay off the critical path
Regulation	Frequency allocation rules	Emission and efficiency standards

Q: Is there a setting where both are optimal?

A: No. There is a **Pareto frontier**: past a point, every improvement in one is paid for out of the other. Design work is choosing where to sit on it, not escaping it.

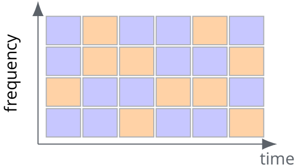
The frontier, drawn once



CRLB = Cramér–Rao Lower Bound: the smallest variance any unbiased estimator can achieve. **Lower is better**, so the good corner is up and to the left.

Every result in the rest of this lecture
is a claim about one point on a curve like this.

The grid is the budget



■ communication RE ■ sensing RE

- One RE is one cell.
- Each carries symbol and power.
- Assign cell, then power.
- Two functions, one grid.

RE = Resource Element, one subcarrier during one OFDM symbol
PSLR = Peak-to-SideLobe Ratio; RoI = Region of Interest in the delay–Doppler plane

Cross-Domain Dual-Functional OFDM Waveform Design, IEEE JSAC 2024

Communication score

Achievable data rate, summed over every RE assigned to communication.

Measured in **time–frequency**: put bits where the channel is good.

Sensing score

PSLR inside the region of interest — how far the true peak stands above the clutter.

Measured in **delay–Doppler**: shape the autocorrelation where targets live.

The same grid is graded twice,
by two examiners in two different domains.

Whoever picks first, wins

Communication-centric

1. Give the good REs to bits.
2. Then tune leftover powers.
3. Sensing takes the remainder.

Rate stays optimal; PSLR is merely good inside the RoI.

Sensing-centric

1. Lock the RoI autocorrelation.
2. Then tune the rest of the plane.
3. Bits take the remainder.

PSLR is locally perfect; rate approaches, not reaches, optimal.

Q: Both designs claim "almost no loss". How can that be?

A: Because the constraint is only enforced inside the **region of interest**. Narrow the room you must be perfect in, and the rest of the grid is free for the other function.

What the simulation actually showed

Communication-centric

Keeps the optimal achievable rate, and still shows good autocorrelation inside the RoI.

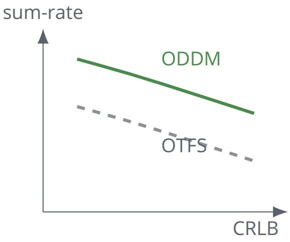
Sensing-centric

Approaches the optimal rate while guaranteeing a locally perfect autocorrelation.

Setup: carrier 240 GHz, subcarrier spacing 240 kHz, 32 consecutive OFDM symbols, 128 subcarriers, symbol length 4.1470 µs, cyclic prefix 1.0368 µs.

> 240 GHz and 240 kHz spacing.
> What kind of link is being assumed here?

Many users, one echo



- Power is split over subcarriers.
- Other users are interference.
- CRLB bounds the estimate.
- The problem splits in two.

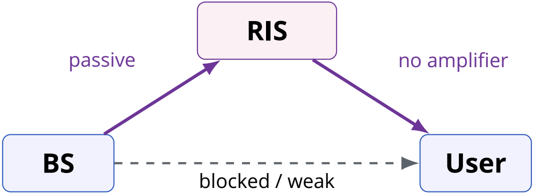
ODDM sits above OTFS across the whole range.

SINR = Signal to Interference-plus-Noise Ratio

Fisher information → CRLB by inversion

Sparser echoes mean less interference,
and less interference is rate you keep.

A surface that edits the channel

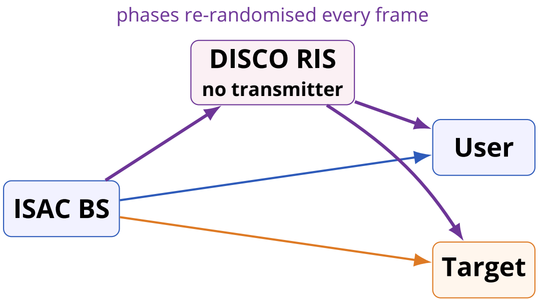


Used honestly it adds coverage, energy efficiency, and a second look at a target.

Used dishonestly it changes your channel without ever emitting a signal of its own.

RIS = Reconfigurable Intelligent Surface   SE / EE = Spectral / Energy Efficiency

The jammer that never transmits



Q: **How do you jam a link with no amplifier and no channel knowledge?**

A: By making the channel move. The base station beamforms on estimated CSI; a surface that reshuffles itself faster than the estimate refreshes turns that CSI stale on arrival. The published name for it is **active channel aging**.

H. Huang et al., ISAC under DISCO physical-layer jamming attacks, IEEE WCL 13(11), 2024

What the attack actually breaks

Without the attack

A well-designed ISAC waveform costs the sum-rate very little. Sensing is nearly free.

With a DISCO RIS

Both functions degrade at once, and the victim has no knowledge of the jammed channel to correct with.

What helps

The damage is statistical, so it can be quantified — and what can be quantified can be budgeted for.

ACA = Active Channel Aging Interference

CSI = Channel State Information

MUI = Multi-User Interference

A passive attacker leaves no emission to detect.
Your only evidence is your own falling SINR.

350 km/h is a hostile radio environment

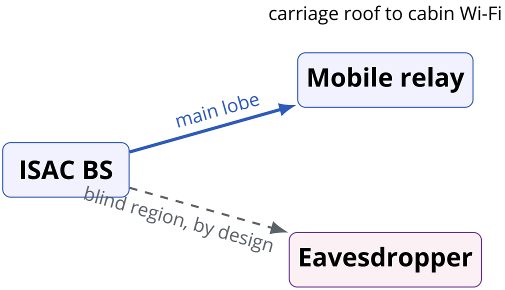
CR400AF high-speed trainset

- mmWave blocks on anything.
- Doppler is large and shifting.
- Handovers every few seconds.
- The track must be watched too.

One system carries passengers and watches the line.

Rooftop relays turn one hard outdoor link
into one easy link plus an indoor problem.

Beamforming with a deliberate blind spot



Two variables

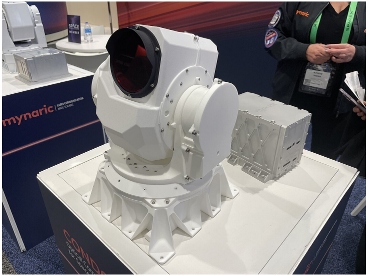
Beamforming matrix, for where the energy points. Waveform, for what the energy says.

Four constraints

Transmit power, constant modulus, sensing SINR, and the blind region.

P. Li et al., Secure High-Speed Train-to-Ground Communications Through ISAC, IEEE IoT J. 11(19), 2024

The same argument, in light



Optical communication terminal, 2025

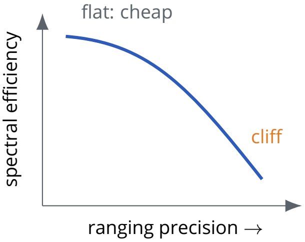
- LiDAR already ranges with light.
- An optical link already carries data.
- Same aperture, detector, and beam.
- No RF licence to argue about.

Q: Why does an optical system need a DC bias?

A: Because intensity cannot be negative. The bias lifts the waveform above zero — and every watt of bias carries no information.

FSO = Free Space Optics   LiDAR = Light Detection and Ranging
IM/DD = Intensity Modulation / Direct Detection   DCO-OFDM = DC-biased Optical OFDM

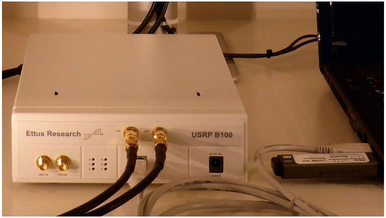
The same curve, drawn in photons



- Tune DC bias with the powers.
- Rate is the objective.
- Precision is the constraint.
- Both metrics saturate.

The right question is never “which is better”.
It is “where on the curve is the cliff”.

Somebody actually built one



USRP software-defined radio (illustrative)

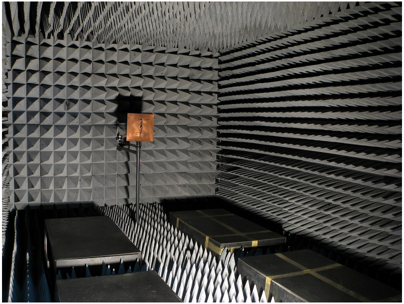
- A host PC drives the bench.
- An FPGA steers the surface.
- USRP plus a converter.
- Transmit on the surface.
- Receive on a horn antenna.
- A target module fakes echoes.

RHS = Reconfigurable Holographic Surface
FPGA = Field-Programmable Gate Array

USRP = Universal Software Radio Peripheral

Two main lobes: one aimed at a user, one at a target.
One aperture, at the same time.

Inside the chamber



RF anechoic chamber, Democritus University of Thrace

- Chamber about 4 m cubed.
- Lobe steered to $-50^\circ, 0^\circ, 20^\circ$.
- Estimated range tracks truth.
- Second lobe carries live data.

Q: Why prove it in an anechoic chamber first?

A: Because the chamber removes every echo you did not create. If sensing fails there, the waveform is wrong; if it fails outdoors, the world is the variable.

IEEE/CIC ICCC 2024, Best Demo Award

Measured numbers are humbler

Range, outdoors

- Target: a metal plate
- Span: 4–21 m
- Accuracy: about 0.425 m

Angle, indoors

- Same metal plate
- Span: -60° to $+60^\circ$
- Accuracy: about 15°

Half a metre and fifteen degrees.
Which applications does that already serve?

The clock, as of this talk

2020–23	2024–26	2027–28	2029–30	2031+
concept and use cases	3GPP Rel-19; trials	spectrum policy	specs and certification	commercial rollout

The 2020–2026 columns describe work that has happened. Everything from 2027 rightward is a forecast made in this talk, not a published schedule.

Standardisation is where the tradeoff stops being a curve and starts being a number somebody must certify.

The whole lecture in one map

	What is optimised	What it costs
Cross-domain OFDM	Resource elements and their powers	Whoever picks second gets the leftovers
Multiuser ODDM	Power per user against the CRLB	Complexity, and hardware that does not exist yet
RIS-ISAC	Beams plus a surface you do not own	An attacker can own the surface instead
HSR ISAC	Beamforming with a deliberate blind spot	Secrecy is bought with coverage
FSO-ISAC	DC bias and subcarrier power	Bias is power that carries nothing

Five settings, one shape.
Every one of them is a point chosen on a frontier.

What is actually unfinished

Settled enough to build

- The tradeoff is understood.
- OFDM-based ISAC works.
- Benches exist and measure.

Still open

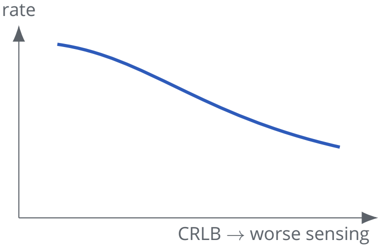
- A killer application.
- Passive-attacker security.
- AI for waveform design.
- Space–air–ground reach.

Q: Which of these is not an engineering problem?

A: The first. A waveform can be optimised; a market cannot. ISAC has a technology looking for the application that pays for it.

SAGIN = Space–Air–Ground Integrated Network

Exit ticket: pick a point on the curve



1. Mark where a self-driving car should sit. Mark where a cabin Wi-Fi link should sit.
2. Name one resource the two designs would allocate differently.
3. Answer in one sentence:
   what does sensing cost the bits in your example?

Appendix: the acronym decoder ring

Area	Term	Meaning / why it matters
System	ISAC	Integrated Sensing and Communication
System	DFRC	Dual-Function Radar Communication; ISAC inside one radio
System	IMT-2030	The ITU framework naming ISAC as a 6G usage scenario
Waveform	OFDM	Orthogonal Frequency Division Multiplexing
Waveform	OTFS	Orthogonal Time Frequency Space; delay–Doppler via ISFFT
Waveform	ODDM	Orthogonal Delay-Doppler Division Multiplexing
Waveform	ISFFT	Inverse Symplectic Finite Fourier Transform
Metric	CRLB	Cramér–Rao Lower Bound; sensing error floor, lower is better
Metric	PSLR	Peak-to-SideLobe Ratio; true peak above the clutter
Metric	SINR	Signal to Interference-plus-Noise Ratio
Metric	PAPR	Peak-to-Average Power Ratio; sets amplifier headroom
Resource	RE / RoI	Resource Element; Region of Interest in delay–Doppler
Surface	RIS / RHS	Reconfigurable Intelligent / Holographic Surface
Attack	DRIS / ACA	DISCO RIS; Active Channel Aging, its jamming mechanism
Optical	FSO, IM/DD	Free Space Optics; Intensity Modulation / Direct Detection

Appendix: the quantities behind every plot

Delay τ range, $R = c\tau/2$

Doppler ν speed, $v = \nu c/(2f_c)$

Bandwidth buys range resolution

Dwell time buys velocity resolution

Rate $\sum \log_2(1 + \mathrm{SINR})$ over REs

CRLB inverse of the Fisher information

PSLR peak over worst sidelobe, in the RoI

Rate is a sum you maximise. CRLB is a floor you cannot go below.

Sensing accuracy is bounded from below.
That is why the frontier bends.

Appendix: which waveform for which problem

Waveform	Best when	Avoid when	Maturity
OFDM	You must ship on existing 5G hardware	Doppler spread is large	Deployed
OTFS	Mobility is high and you want an OFDM front end	Sidelobes must be clean	Prototypes
ODDM	Sensing accuracy and multiuser separation both matter	You need it this year	Research
DCO-OFDM	The medium is light, not radio	Illumination power is scarce	Research

Q: Why would anyone still choose OFDM for sensing?

A: Because the alternative costs a new modem in every device. A worse waveform that exists beats a better one that does not.

Sources and further reading

- **Zhu Han (Univ. of Houston), "Integrated Sensing and Communication for 6G"**
  The invited talk this lecture is built from. Slides and code: wireless.egr.uh.edu.
- **ITU-R M.2160: IMT-2030 framework**
  Where ISAC becomes an official 6G usage scenario.
- **Cross-domain dual-functional OFDM waveform design**
  IEEE JSAC 2024. Source of the RE-allocation results.
- **H. Huang et al., ISAC under DISCO physical-layer jamming attacks**
  IEEE Wireless Commun. Lett. 13(11), 3044–3048, Nov. 2024.
- **P. Li et al., Secure high-speed train-to-ground communications through ISAC**
  IEEE Internet of Things J. 11(19), 31235–31248, Oct. 2024.
- **R. Liu et al., SNR/CRB-constrained beamforming for RIS-ISAC**
  IEEE Trans. Wireless Commun. 23(7), 7456–7470, Jul. 2024.